

Homework 2: Existence, uniqueness, scalar potentials, and numerical error

MATH 515 — Fall 2026

Due Tuesday, September 22, 2026

Assignment convention. Each numbered problem reproduces the full assigned statement from Steven H. Strogatz, *Nonlinear Dynamics and Chaos*, third edition (the specific section/problem number is given parenthetically), with some mild editing for clarity. Any additional requirements are labeled **Course-specific instructions** and are part of the assignment.

Problem 1: A general nonuniqueness threshold [Challenge]

Textbook problem (Strogatz, third edition, Exercise 2.5.5, “A general example of non-uniqueness”):

Consider the initial value problem

$$\dot{x} = |x|^{p/q}, \quad x(0) = 0,$$

where p and q are positive integers with no common factors.

- (a) Show that there are an infinite number of solutions for $x(t)$ if $p < q$.
- (b) Show that there is a unique solution if $p > q$.

Course-specific instructions. Complete both parts. In part (a), first construct a solution that stays at zero until a chosen time $\tau \geq 0$ and then becomes positive. Then describe all solutions defined for every real t , including a possible negative branch before a time $\sigma \leq 0$. Check the differential equation at the joining times, and explain why your list is complete. In part (b), identify the hypothesis that guarantees uniqueness and verify it for this vector field.

Problem 2: Why scalar autonomous flows cannot oscillate

Textbook problem (Strogatz, third edition, Exercise 2.6.1 and 2.6.2, “Impossibility of oscillations”):

Exercise 2.6.1. Explain this paradox: a simple harmonic oscillator $m\ddot{x} = -kx$ is a system that oscillates in one dimension (along the x -axis). But the text says one-dimensional systems cannot oscillate.

Exercise 2.6.2 (No periodic solutions to $\dot{x} = f(x)$). Here is an analytic proof that periodic solutions are impossible for a vector field on a line. Suppose on the contrary that $x(t)$ is a nonconstant periodic solution, so $x(t) = x(t + T)$ for every t and some $T > 0$. Derive a contradiction by considering

$$\int_t^{t+T} f(x) \frac{dx}{dt} dt.$$

Course-specific instructions. Treat the two exercises as one problem. Resolve the apparent harmonic-oscillator paradox first, and then give the integral contradiction requested in Exercise 2.6.2. Assume f is continuous on an open interval containing the solution's values. Explain why the same integral is zero by periodicity and positive if the solution is nonconstant.

(Note: The statement of Exercise 2.6.2 given here is slightly different from the textbook's wording. The textbook adds a minimality assumption to the period of the oscillation, which is not actually necessary or, in fact, sensible: consider, for instance, a constant solution, which has every positive number as a period. So, to keep things simple, it was removed.)

Problem 3: A parameter-dependent potential

Textbook problem (Strogatz, third edition, Exercise 2.7.6):

For the following vector field, plot the potential function $V(x)$ and identify all the equilibrium points and their stability:

$$\dot{x} = r + x - x^3,$$

for various values of r .

Course-specific instructions. Complete the exercise for all real r . Locate the two critical parameter values exactly, classify every equilibrium in every regime, and show how the potential and phase line encode the same bifurcations.

Problem 4: An analytically intractable initial-value problem [Challenge]

Textbook problem (Strogatz, third edition, Exercise 2.8.6, “Analytically intractable problem”):

Consider the initial value problem

$$\dot{x} = x + e^{-x}, \quad x(0) = 0.$$

In contrast to Exercise 2.8.3, this problem cannot be solved analytically.

- (a) Sketch the solution $x(t)$ for $t \geq 0$.
- (b) Using some analytical arguments, obtain rigorous bounds on the value of x at $t = 1$. In other words, prove that $a < x(1) < b$, for a, b to be determined. By being clever, try to make a and b as close together as possible. (Hint: Bound the given vector field by approximate vector fields that can be integrated analytically.)
- (c) Now for the numerical part: Using the Euler method, compute x at $t = 1$, correct to three decimal places. How small does the stepsize need to be to obtain the desired accuracy? (Give the order of magnitude, not the exact number.)
- (d) Repeat part (c), now using the Runge–Kutta method. Compare the results for stepsizes $\Delta t = 1$, $\Delta t = 0.1$, and $\Delta t = 0.01$.

Course-specific instructions. Complete all parts; use the classical fourth-order Runge–Kutta method in part (d). For part (b), prove a nontrivial lower and upper bound; you need not find the best possible bounds. For parts (c)–(d), submit code or precise pseudocode and a table using at least three successive step sizes $h, h/2, h/4$ for each method. Use the resulting table to estimate the error and check that further refinement preserves your answer rounded to three decimal places. State which conclusions are supported

by numerical evidence and which are proved by analytic inequalities. Let X_h denote the approximation obtained with step size h . For three successive step sizes $h, h/2, h/4$, use

$$p_{\text{obs}} = \log_2 \left(\frac{|X_h - X_{h/2}|}{|X_{h/2} - X_{h/4}|} \right)$$

where p_{obs} is the *observed order of convergence*; i.e., it estimates how the numerical error scales with the step size, without requiring the exact solution, so that an observed order near p suggests that halving the step size reduces the error by approximately a factor of 2^p . If the differences vanish in machine arithmetic, use coarser steps to estimate the order. A certified error bound for the numerical methods is not required. **Computational submission.** Include code that would allow another person to reproduce the calculation or plot. Clearly label every reported numerical value and plot, and explain what the output shows.

Problem 5: Local truncation error and method order

Textbook problem (Strogatz, third edition, Exercise 2.8.7 and 2.8.8, “Error estimates”):

Exercise 2.8.7 (Error estimate for Euler method). In this question you will use Taylor series expansions to estimate the error in taking one step by the Euler method. The exact solution and the Euler approximation both start at $x = x_0$ when $t = t_0$. We want to compare the exact value $x(t_1) \equiv x(t_0 + \Delta t)$ with the Euler approximation $x_1 = x_0 + f(x_0)\Delta t$.

- Expand $x(t_1) = x(t_0 + \Delta t)$ as a Taylor series in Δt , through terms of $O(\Delta t^2)$. Express your answer solely in terms of x_0 , Δt , and f and its derivatives at x_0 .
- Show that the local error $|x(t_1) - x_1| \sim C\Delta t^2$ and give an explicit expression for the constant C . (Generally one is more interested in the global error incurred after integrating over a time interval of fixed length $T = n\Delta t$. Since each step produces an $O(\Delta t^2)$ error, and we take $n = T/\Delta t = O(\Delta t^{-1})$ steps, the global error $|x(t_n) - x_n|$ is $O(\Delta t)$, as claimed in the text.)

Exercise 2.8.8 (Error estimate for the improved Euler method). Use the Taylor series arguments of Exercise 2.8.7 to show that the local error for the improved Euler method is $O(\Delta t^3)$.

Course-specific instructions. Treat the two exercises as a single comparison problem. Following the course notes, call the one-step difference obtained when the numerical step starts from the exact value the *local defect*. Derive its leading term for Euler and improved Euler directly from Taylor expansions, and explain how these local orders lead to the corresponding global orders under the explicit one-step bound below. (The textbook statement above calls the same one-step quantity the “local error”.) The textbook’s asymptotic-equivalence claim for Euler applies only when the leading coefficient is nonzero; otherwise state the leading expansion and the resulting $O(h^2)$ bound without asserting asymptotic equivalence to a nonzero multiple of h^2 . Assume that $f \in C^3$ on a neighborhood of all exact and numerical states under consideration. For the global-error step, let $\Psi_h(t, u)$ denote one numerical step of size h beginning from value u at time t , and assume on the relevant region that

$$|\Psi_h(t, u) - \Psi_h(t, v)| \leq (1 + Lh)|u - v|$$

for constants $L \geq 0$ and $h_0 > 0$, independent of h , whenever $0 < h \leq h_0$. Use this bound to explain why an $O(h^{p+1})$ local defect produces an $O(h^p)$ error over a fixed interval $[t_0, t_0 + T]$, where $T > 0$ is independent of h . Here the local-defect bound must hold uniformly over every step through $t_0 + T$, with its constant independent of the step index and of h .